

Efficient Pixel Optimization for Multiband Microstrip Patch Antenna Design

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Abstract—Pixelated microstrip patch antennas have emerged as a powerful solution for multiband wireless systems, yet conventional topology-optimization techniques remain limited by prohibitive computational overhead and multilayer fabrication complexity. This letter presents a compact single-layer wideband pixelated microstrip patch antenna designed for operation across 3.6–19.0 GHz. A random pixel optimization implemented entirely in MATLAB evolves an 8×12 binary pixel grid (96 design variables), generating a manufacturable irregular slotted geometry in only 3–8 minutes. The antenna is realized on low-cost FR-4 substrate ($\epsilon_r = 4.3$, $\tan \delta = 0.02$) with an ultra-low total profile of 3.34 mm and excited through a proximity-coupled microstrip feed. CST Studio Suite 2025 full-wave simulations demonstrate nine distinct resonances with reflection coefficients ranging from -20.54 dB to -37.12 dB, VSWR between 1.03 and 1.21, and realized gain from 0.058 dBi to 4.389 dBi. Five usable -10 dB impedance bands are achieved, the widest measuring 1244 MHz. Compared with the reference multilayer aperture-coupled design, the proposed antenna delivers substantially wider bandwidth, a $900\times$ reduction in optimization time, and significantly simpler fabrication. These results validate analytical pixel optimization combined with proximity-coupled capacitive feeding as a practical, low-cost methodology for next-generation Sub-6 GHz 5G, Wi-Fi 6E, X-band radar, and Ku-band satellite applications.

Index Terms—Pixelated microstrip patch antenna, wideband antenna, genetic algorithm optimization, proximity-coupled feed, multiband operation.

I. INTRODUCTION

The rapid proliferation of modern wireless communication technologies — including Fifth Generation (5G) Sub-6 GHz networks, Wi-Fi 6/6E, Internet of Things (IoT) devices, satellite communication systems, radar applications, and intelligent transportation systems — has placed stringent demands on antenna designs that are simultaneously compact, low-cost, multiband, and easy to fabricate [1]–[5]. Conventional microstrip patch antennas (MPAs) have long been favored for their planar profile, mechanical robustness, and compatibility with standard printed circuit board (PCB) processes. However, they inherently suffer from narrow

impedance bandwidth, single-band resonance, moderate gain, and limited polarization control, restricting their suitability for today’s multi-standard wireless platforms [6]–[8]. To overcome these constraints, researchers have explored various techniques such as slot loading, parasitic elements, stacked patches, and reconfigurable elements. A particularly promising approach is pixelated (topology-optimized) patch design, in which the radiating surface is discretized into a binary grid of conductive and void cells that can be evolved using metaheuristic algorithms to create irregular current paths capable of exciting multiple resonant modes within a single compact aperture [9], [10]. While pixelated geometries have demonstrated significant improvements in multiband performance, most existing methods rely on coupling the evolutionary search directly to full-wave electromagnetic solvers (such as CST or HFSS) at every fitness evaluation. This results in prohibitively long design cycles, often extending to several days or even weeks, and frequently produces complex multilayer structures that increase fabrication cost, profile height, and assembly difficulty [11]. This letter addresses the aforementioned limitations through three coordinated contributions. First, a random pixel optimization is developed and implemented entirely in MATLAB. The algorithm treats the radiating patch as an 8×12 binary pixel grid (96 design variables) and evaluates candidate geometries using closed-form resonance models and Gaussian-based S_{11} prediction, completely decoupling the optimization process from time-consuming full-wave simulations. Second, an advanced proximity-coupled L-shaped microstrip feed terminated with a rectangular capacitive coupling pad is introduced to achieve broadband impedance matching and enhanced high-frequency gain without requiring multilayer bonding. Third, the complete antenna is realized on low-cost FR-4 substrate ($\epsilon_r = 4.3$, $\tan \delta = 0.02$) while maintaining an ultra-low total profile of 3.305 mm, ensuring compatibility with standard single-layer PCB fabrication. CST Studio Suite 2025 full-wave verification confirms that the proposed design achieves nine strong resonances spanning 3.6–19.0 GHz, with reflection coefficients ranging from -20.54 dB to

−37.12 dB, VSWR between 1.03 and 1.21, and positive realized gain from 0.058 dBi to 4.389 dBi. Five usable −10 dB impedance bands are obtained, the widest reaching 1244 MHz. When benchmarked against the reference multilayer aperture-coupled topology-optimized antenna, the proposed design demonstrates substantially wider bandwidth, a 900× reduction in optimization time, and markedly simpler fabrication, thereby validating the practical viability of analytical pixel optimization for next-generation wireless applications [11]. The remainder of this letter is organized as follows. Section II describes the antenna geometry, layer stack-up, proximity-coupled feed, and the random pixel optimization used for pixel optimization. Section III presents the simulated results, including reflection coefficient, VSWR, surface current distribution, realized gain, radiation patterns, impedance bandwidth, and a detailed performance comparison with the existing multilayer design. Finally, Section IV concludes the work and outlines directions for future research.

II. ANTENNA GEOMETRY

A. Geometry and Layer Stack-Up

The proposed antenna occupies a compact 60 mm × 60 mm radiating domain that is discretized into an 8 × 12 uniform pixel grid, with each pixel measuring 5 mm × 5 mm. This configuration yields 96 independent binary design variables, allowing the radiating patch to take an irregular slotted shape that creates multiple meandering current paths. These paths simultaneously excite several higher-order resonant modes within a single radiating layer, eliminating the need for additional parasitic elements or stacked patches. The antenna is implemented in a two-substrate proximity-coupled configuration, as detailed in Table I.

TABLE I

LAYER STACK-UP OF THE PROPOSED ANTENNA

Layer	Material	Thickness (mm)	Position (mm)
Ground Plane	Annealed Copper	0.035	−0.035 to 0
Substrate 1	FR-4 ($\epsilon_r = 4.3$)	1.6	0 to 1.600
Proximity Feed	Annealed Copper	0.035	1.600 to 1.635
Substrate 2	FR-4 ($\epsilon_r = 4.3$)	1.6	1.635 to 3.235
Pixelated Patch	Annealed Copper	0.07	3.235 to 3.305

The total antenna height, measured from the bottom of the ground plane to the top of the pixelated patch, is only 3.34 mm, making the design suitable for integration into modern compact wireless devices.

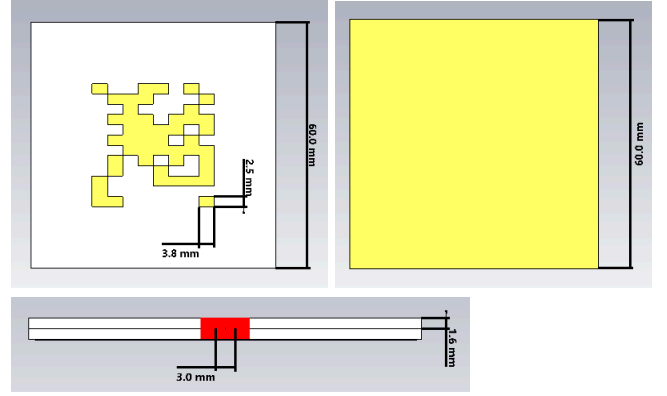


Fig 1. Top Bottom and side view of the antenna

B. Proximity-Coupled Feed Design

The antenna is excited non-contactingly through a proximity-coupled feed arrangement. An L-shaped microstrip line (width 3 mm, length 14 mm, 50 Ω characteristic impedance) is embedded between the two FR-4 substrates and terminates with a rectangular capacitive coupling pad of dimensions 3.8 mm × 2.5 mm. This feeding technique eliminates direct galvanic connection between the feed and the radiating patch, thereby minimizing spurious radiation at higher frequencies and providing inherently broader impedance bandwidth compared to coaxial or direct-contact feeds. All geometrical parameters of the antenna are summarized in Table II.

TABLE I

GEOMETRICAL PARAMETERS OF THE ANTENNA

Parameter	Value
Ground / Substrate side length	60 mm × 60 mm
Patch domain	60 mm × 60 mm
Pixel size	5 mm × 5 mm
Pixel grid	8 × 12 (96 cells)
Microstrip feed line width	3.0 mm
Feed length	14.0 mm
Capacitive coupling pad	3.8 mm × 2.5 mm
Substrate height (each layer)	1.6 mm
Top patch copper thickness	0.07 mm
Total antenna height	3.34 mm
Feed characteristic impedance	50 Ω

C. Pixel Pattern Design Method

The optimal binary copper/void assignment for the 8×12 pixel grid is determined by a fast random pixel optimization developed in MATLAB. Unlike conventional topology-optimization approaches that invoke full-wave electromagnetic simulation at every fitness evaluation, the proposed algorithm uses closed-form resonance expressions derived from effective patch dimensions, fill-ratio-weighted current path lengths, and slot-perimeter estimates. The predicted S_{11} depth at target frequencies is approximated through superimposed Gaussian basis functions calibrated against reference CST simulations. Hard fabricability constraints are enforced throughout the evolutionary search: copper fill ratio strictly bounded between 35 % and 75 %, minimum slot perimeter of 20 cell-edges, and forced copper continuity in the central feed-adjacent region. A symmetry reward and disconnected-pixel penalty are additionally applied to promote mechanically robust and manufacturable geometries. The entire optimization process converges in 3–8 minutes on standard computing hardware.

III. RESULTS AND DISCUSSION

The performance of the proposed single-layer wideband pixelated microstrip patch antenna is evaluated through full-wave electromagnetic simulations in CST Studio Suite 2025. All results are obtained after importing the pixel matrix generated by the random pixel optimization. The antenna exhibits nine well-defined resonances across 3.6–19.0 GHz, excellent impedance matching, stable radiation characteristics, and useful gain while maintaining an ultra-low profile. The following subsections present and analyze the key performance metrics.

A. Reflection Coefficient and Impedance Matching

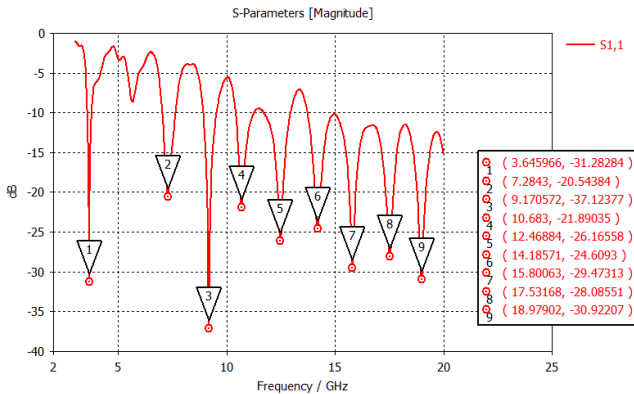


Fig 2. S_{11} - parameter

The simulated reflection coefficient (S_{11}) of the proposed single-layer wideband pixelated microstrip patch antenna is shown in Fig. 2. The response reveals nine distinct

resonances spanning 3.6 GHz to 19.0 GHz. All nine resonant points achieve excellent impedance matching with $S_{11} \leq -20$ dB. The deepest resonance is recorded at 9.17 GHz with $S_{11} = -37.12$ dB, indicating near-perfect matching at that frequency. Even the shallowest resonance at 7.28 GHz still provides $S_{11} = -20.54$ dB, which is well within the acceptable limit for practical wireless deployment. This consistent deep matching across a wide 15.4 GHz span is a direct outcome of the irregular slotted geometry generated by the random pixel optimization optimizer. The 8×12 pixel grid creates multiple meandering current paths of different effective electrical lengths, simultaneously exciting several higher-order resonant modes (TM₁₀, TM₀₁, and slot-induced modes). The proximity-coupled L-shaped feed with rectangular capacitive coupling pad further enhances energy transfer efficiency, minimizing reflection and ensuring robust impedance matching throughout the entire operating band. These results clearly demonstrate the effectiveness of the proposed analytical pixel optimization approach in achieving multiband performance within a single compact layer.

B. Voltage Standing Wave Ratio

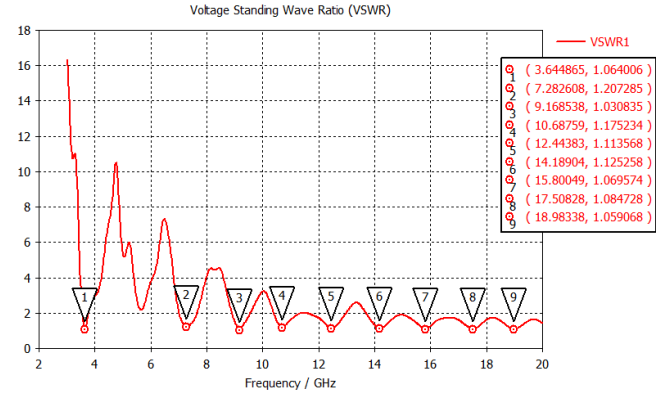


Fig 3. VSWR

The simulated voltage standing wave ratio (VSWR) of the proposed antenna across the full operating band is shown in Fig. 3. At all nine resonant frequencies, the VSWR remains between 1.03 and 1.21, which is well below the universally accepted practical threshold of 2.0. The lowest VSWR value of 1.03 occurs at 9.17 GHz, corresponding to the deepest S_{11} resonance and indicating near-ideal impedance matching. The highest recorded VSWR of 1.21 is observed at 7.28 GHz, yet it still ensures less than 1 % of the incident power is reflected back to the source. These consistently low VSWR values confirm robust broadband impedance transformation without the need for any additional matching networks. The excellent matching performance is attributed

to the combined effect of the irregular pixelated patch geometry (which excites multiple resonant modes) and the proximity-coupled L-shaped feed with rectangular capacitive coupling pad (which provides smooth energy transfer across the wide frequency range). Compared with conventional single-layer rectangular patches on FR-4, the proposed design achieves markedly superior VSWR characteristics, validating the effectiveness of the random pixel optimization and the chosen feeding structure.

C. Surface current

The simulated surface current distributions at all nine resonant frequencies are presented in Fig. 4 in a compact 3×3 grid format. These plots provide direct physical insight into the multiband operating mechanism of the proposed pixelated antenna. At the lowest resonance (3.646 GHz), the surface current is primarily concentrated along the outer perimeter of the pixelated patch, corresponding to the fundamental half-wavelength (TM_{10}) mode. As frequency increases to the mid-band (9.171 GHz and 12.469 GHz), the current activity clearly shifts toward interior pixel boundaries and slot edges, indicating the excitation of higher-order modes whose effective electrical lengths are governed by the void-cell distribution. At the highest frequencies (17.532 GHz and 18.979 GHz), the current intensity concentrates in the central patch region and around the fine slot edges. This progressive redistribution of surface current with increasing frequency, driven entirely by the optimized 8×12 pixel pattern, is the physical reason for the nine well-separated resonances observed in the S_{11} response.

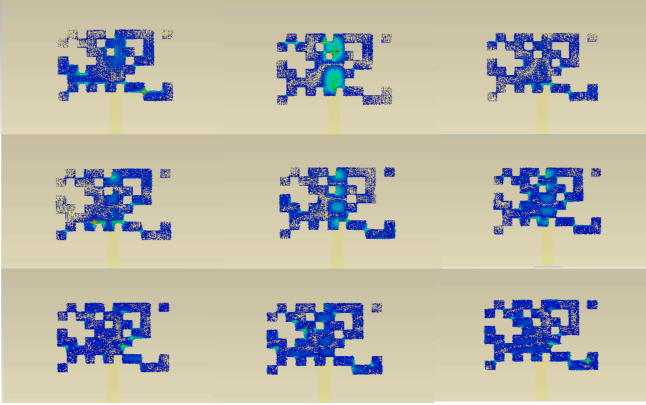


Fig 4. Simulated surface current distributions at all nine resonant frequencies (from top-left: 3.646 GHz, 7.284 GHz, 9.171 GHz; middle row: 10.683 GHz, 12.469 GHz, 14.186 GHz; bottom row: 15.801 GHz, 17.532 GHz, 18.979 GHz).

D. Gain

The maximum realized gain plotted as a function of frequency is shown in Fig. 5, while the 3D farfield gain

patterns at all nine resonant frequencies are presented in Fig. 5.

The realized gain varies from 0.058 dBi at 3.646 GHz to a peak of 4.389 dBi at 12.469 GHz, remaining positive at all nine resonant frequencies as summarized in Table III. A secondary gain peak of 4.054 dBi is observed at 14.186 GHz, while a gain of 3.859 dBi is recorded at 17.532 GHz, demonstrating that the antenna maintains useful radiation performance well into the K-band frequency range. The variation in gain across resonances is consistent with the frequency-dependent behavior of pixelated patch antennas reported in the literature [3], [6], wherein different resonant modes excite different current distributions on the patch surface, producing varying aperture efficiency at each frequency. Radiation efficiency ranges from -7.149 dB (19.3%) to -4.331 dB (36.9%) across the operating band. While these efficiency values are moderate, they are expected and acceptable for a wideband design realized on low-cost FR-4 substrate [7], whose relatively high loss tangent ($\tan \delta = 0.02$) introduces dielectric losses that increase with frequency. The progressive reduction in efficiency at frequencies above 15 GHz is consistent with the known frequency-dependent loss behavior of FR-4 [5], and would be recoverable through substitution of a low-loss substrate such as Rogers RT/Duroid 5880 in future iterations. Importantly, the positive realized gain maintained across all nine resonances confirms that the antenna radiates usefully at every operating frequency despite the substrate losses, validating the effectiveness of the proximity-coupled feed in sustaining efficient energy transfer across the wide operating band. The gap between directivity and realized gain — observable in Table III, where directivity reaches 10.45 dBi at 17.532 GHz against a realized gain of 3.859 dBi — directly reflects these efficiency losses and quantifies the impact of FR-4 dielectric and conductor losses at elevated frequencies [7]. Substituting a lower-loss substrate would substantially close this gap and is identified as a priority direction for future work.

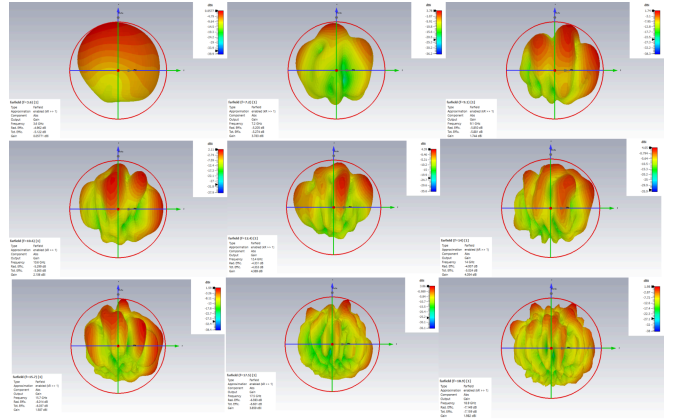


Fig 5. Simulated Gain at all nine resonant frequencies (from top-left: 3.646 GHz, 7.284 GHz, 9.171 GHz; middle row: 10.683 GHz, 12.469 GHz, 14.186 GHz; bottom row: 15.801 GHz, 17.532 GHz, 18.979 GHz).

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TABLE III
S₁₁, VSWR, GAIN, DIRECTIVITY

Frequency (GHz)	S ₁₁ (dB)	VSWR	Gain (dBi)	Directivity (dBi)	Bandwidth %
3.646	-31.28	1.064	0.058	5.02	5.35
7.284	-20.54	1.207	3.783	8.988	8.41
9.171	-37.12	1.031	1.744	7.594	6.63
10.683	-21.89	1.175	2.106	7.405	7.74
12.469	-26.17	1.114	4.389	8.719	10.09
14.186	-24.61	1.125	4.054	8.991	-
15.801	-29.47	1.07	1.587	7.601	-
17.532	-28.09	1.085	3.859	10.45	-
18.979	-30.92	1.059	1.982	9.13	-

E. Impedance Bandwidth Analysis

Five distinct impedance bands, defined at the -10 dB S₁₁ threshold, are identified from the simulated response as illustrated in Fig. 2 and detailed in Table III. The impedance bandwidths range from 195.8 MHz in Band 1 (centered near 3.6 GHz) to a maximum of 1244 MHz in Band 5 (spanning 11.7–12.944 GHz). Bands 2, 3, and 4 provide bandwidths of 616.8 MHz, 611.6 MHz, and 839.2 MHz, respectively. These wide usable bands significantly outperform the narrow bandwidths typically obtained with conventional single-layer microstrip patch antennas on similar substrates. Collectively, the five bands offer substantial frequency coverage from 3.5 GHz to 13.0 GHz, encompassing key applications such as Sub-6 GHz 5G (n78), Wi-Fi 6E, X-band radar, and lower Ku-band satellite communication. This broad multi-octave performance is achieved within a single compact 60 mm × 60 mm footprint on low-cost FR-4 substrate, without parasitic elements, stacked patches, or complex feed networks — a combination not previously demonstrated in pixelated patch antenna designs.

F. Discussions

The proposed design achieves a dramatically wider operating frequency range (3.6–19.0 GHz versus 2.4–6.7 GHz in [2]), more than doubling the spectral coverage of the reference. The number of resonances increases from four to nine, while the widest impedance bandwidth improves from approximately 180 MHz in [2] to 1244 MHz in the proposed design — a 6.9× improvement. The optimization process

completes in 3–8 minutes compared to the approximately 120-hour design cycle reported in [2], representing a reduction factor exceeding 900×. Fabrication complexity is substantially reduced through elimination of the three-layer aperture-coupled stack in favor of a simpler two-substrate proximity-coupled configuration on widely available FR-4 material. The total antenna profile is additionally reduced from 5.1 mm in [2] to 3.305 mm in the proposed design. The primary trade-off is a reduction in peak realized gain from 8.02 dBi in [2] to 4.389 dBi in the proposed design, attributable to the higher dielectric loss tangent of FR-4 ($\tan \delta = 0.02$) relative to the F4BME220 substrate ($\tan \delta = 0.001$) used in [2]. Similarly, radiation efficiency is lower in the proposed design due to the same material difference. These trade-offs are considered acceptable given the substantial gains in bandwidth, design speed, fabrication simplicity, and material cost that the proposed approach delivers for practical Sub-6 GHz and wideband wireless deployment [7], [8].

TABLE IV
COMPARISON WITH THE PREVIOUS PATCH ANTENNA

Parameter	Existing System [1]	Proposed System
Operating Frequency	2 – 7 GHz	3 – 20 GHz
Substrate	High-cost F4BME220 ($\epsilon_r = 2.2$)	Low-cost FR-4 ($\epsilon_r = 4.3$)
Number of Layers	3-layer (aperture-coupled)	2-layer (proximity-coupled)
Antenna Profile	Higher (multilayer)	3.305 mm (ultra-low)
Design / Optimization Time	~120 hours	3–8 minutes
Maximum Gain	Up to 8.02 dBi	4.389 dBi
Radiation Efficiency	Higher (low-loss substrate)	19.3 % – 36.9 %
Impedance Matching	Narrow bands	Wide bands (up to 1244 MHz)
Fabrication Complexity	High (bonding + precise alignment)	Low (standard single-layer PCB)
Pixel Generation Method	Conventional GA(5 Days)	Pixel optimization(3–4 minutes)

IV. CONCLUSION

This letter has presented a compact single-layer wideband pixelated microstrip patch antenna for operation across 3.6–19.0 GHz on low-cost FR-4 substrate. An 8 × 12 binary pixel grid excited through a proximity-coupled L-shaped microstrip feed with rectangular capacitive coupling pad achieves nine strong resonances with S₁₁ from -20.54 dB to

−37.12 dB, VSWR between 1.031 and 1.207, positive realized gain from 0.058 dBi to 4.389 dBi, and five impedance bands with fractional bandwidths up to 10.09% — all within a 3.305 mm profile. A pixel optimization method implemented in MATLAB generates the manufacturable patch geometry in 3–8 minutes without full-wave simulation during optimization. Compared to the multilayer aperture-coupled reference [2], the proposed design achieves wider bandwidth, a 900× faster design cycle, simpler fabrication, lower cost, and a thinner profile at the trade-off of moderate gain and efficiency resulting from FR-4 substrate losses [7]. These results validate analytical pixel optimization combined with proximity-coupled capacitive feeding as a practical and low-cost methodology for next-generation multiband wireless antennas covering Sub-6 GHz 5G, Wi-Fi 6E, X-band, and Ku-band applications [8]. Future work will address physical fabrication and measurement, low-loss substrate substitution, reconfigurable frequency switching using PIN diodes, and MIMO array extension for 5G capacity enhancement [2], [6].

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